

1. Description and Motivation:

Customer churn prediction is a key problem in banking and telecommunications, where identifying customers likely to leave enables targeted retention strategies. This project formulates churn prediction as a supervised binary classification task and compares two core machine learning models taught in this module: Naive Bayes and Decision Trees.

Aim: build a rigorous evaluation pipeline (baseline + cross-validation + hold-out testing) and provide a fair comparative analysis of the two models.

2. Dataset and Exploratory Analysis:

- We used the Telco Customer Churn dataset from Kaggle [1], containing demographic, service usage, and billing data for 7,043 customers. Churn is a binary target (Yes/No).
- Exploratory analysis showed a class imbalance with 73.5% non-churners and 26.5% churners (Figure 1) motivating the use of metrics beyond accuracy , such as F1-score and ROC–AUC.
- Data preparation included removing customer identifiers, converting text attributes to categorical variables, and cleaning TotalCharges to a numeric format. Rows with missing values were removed.
- Simple, interpretable feature engineering was applied. An average monthly spend feature was created to better capture customer value over time, and a binary long-term contract indicator was introduced to reflect contractual commitment.

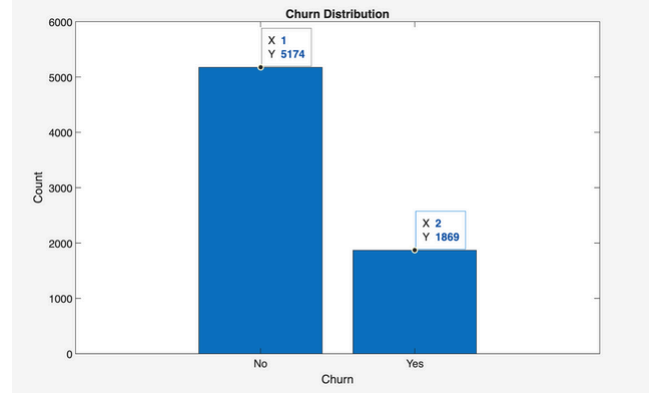


Figure 1: Churn Distribution

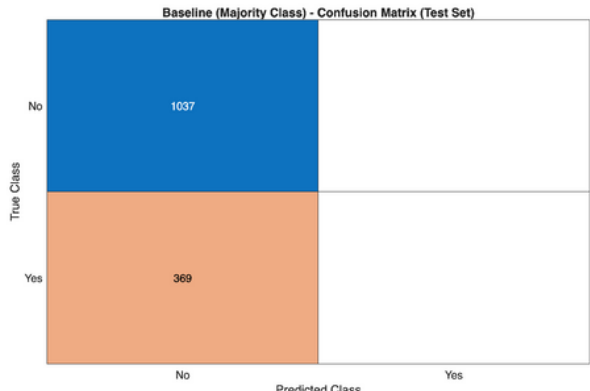


Figure 2: Baseline confusion matrix

3. Hypothesis Statement

We hypothesised that both Naive Bayes and Decision Trees would outperform a simple baseline classifier.

We further expected Naive Bayes to achieve higher recall due to its probabilistic nature, while a tuned Decision Tree would provide better overall discrimination and accuracy.

4. Baseline Model

- A baseline was included to provide a point of reference, answering:
- “Do the machine learning models outperform a trivial strategy?”
- The baseline predicts the majority class (“No”) for all test samples, achieving a test accuracy of 0.738 due to dataset imbalance. However, it fails to identify any churners, predicting that no customers leave.
- As shown in Figure 2, the empty predicted Yes column is expected and highlights why accuracy alone is misleading, motivating the use of metrics such as F1-score and ROC–AUC.

5. Models Considered

Naive Bayes

Naive Bayes is a probabilistic classifier based on Bayes’ theorem, which assumes conditional independence between features given the class label.

Pros:

- We used Naive Bayes as a fast and interpretable probabilistic baseline
- It provides class probabilities, allowing threshold-independent evaluation using ROC curves
- Its simplicity makes it a useful reference point for assessing more flexible models

Limitations observed:

- The independence assumption is unlikely to hold for correlated customer behaviour features
- Performance can be affected by feature distributions, motivating careful preprocessing

Decision Tree

Decision Trees learn a set of hierarchical decision rules by recursively splitting the feature space.

Pros:

- The model produces interpretable rules, which are valuable in customer-facing domains
- It naturally captures non-linear relationships and feature interactions
- It handles categorical variables well without strong distributional assumptions

Limitations observed:

- Trees can overfit if allowed to grow too deep
- Model performance is highly sensitive to hyperparameters, motivating cross-validation and tuning

5. Methodology

We designed the experimental pipeline as follows:

- Data split into 80% training and 20% held-out test set
- 5-fold cross-validation used on the training set for model selection (see Figure 3)
- Hyperparameter tuning applied to the Decision Tree (see Figure 4)
- Final evaluation performed only on the test set
- Metrics used: Accuracy, F1-score, ROC-AUC
- This ensured a fair comparison and avoided information leakage.

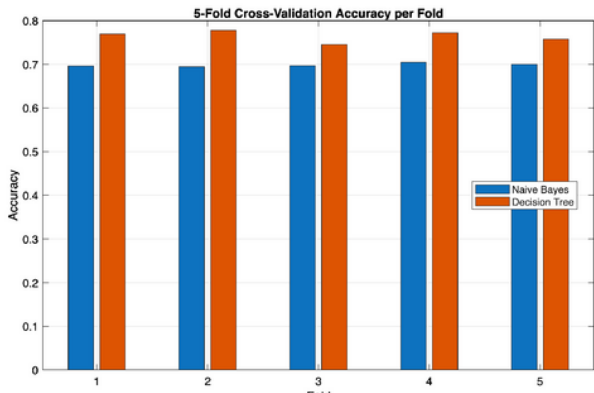


Figure 3: Cross-validation accuracy per fold



Figure 4: Decision Tree Tuning (Cv Mean AUC)

6. Cross-Validation Results

Using 5-fold cross-validation on the training set, we obtained the following results:

Model	Mean Accuracy	Std	Mean AUC	Std
Naive Bayes	0.699	0.004	0.818	0.01
Decision Tree	0.765	0.013	0.808	0.02

Decision Trees achieved higher mean accuracy, while Naive Bayes showed slightly more stable AUC across folds.

7. Hyperparameter Tuning

Hyperparameter tuning was applied only to the Decision Tree, as its performance is highly sensitive to model complexity parameters such as tree depth and leaf size. Naive Bayes has a fixed model structure with limited tunable parameters, and tuning typically provides minimal performance gains.

A grid search was conducted using cross-validation over:

- MinLeafSize: {1, 2, 5, 10, 20, 50}
- MaxNumSplits: {10, 25, 50, 100}

The optimal configuration was selected based on mean cross-validated AUC.

Best parameters: MinLeafSize = 2, MaxNumSplits = 50, Mean CV AUC = 0.826

8. Experimental Results

We evaluated all models on a held-out test set (20%) that was not used during training, cross-validation, or hyperparameter tuning. Performance is reported using accuracy, precision, recall, F1-score, and AUC to capture both overall performance and class-specific behaviour in this imbalanced dataset.

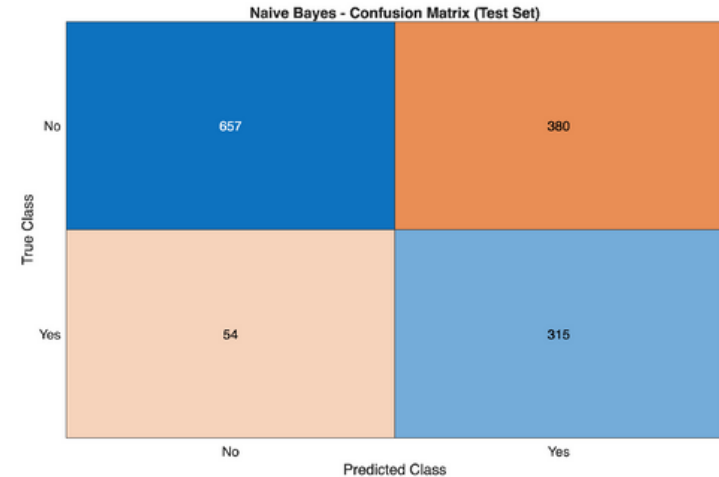


Figure 5: Naive Bayes Confusion Matrix

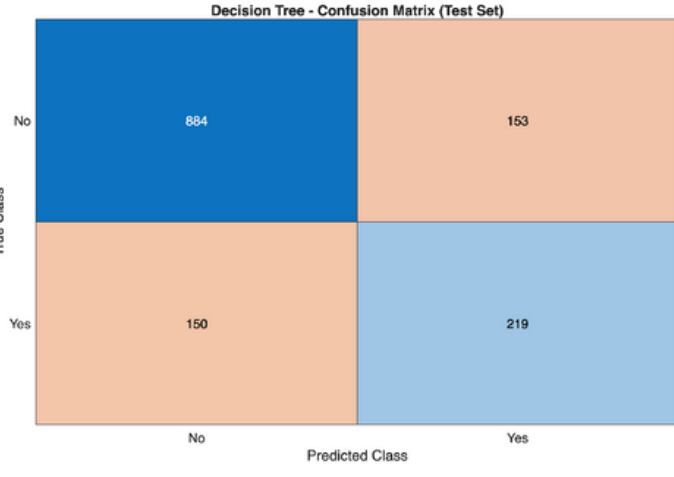


Figure 6: Decision tree Confusion Matrix

Key Observations (Figure 5 & 6)

- Naive Bayes achieves very high recall, correctly identifying most churners, but at the cost of lower precision and overall accuracy.
- The tuned Decision Tree provides a more balanced trade-off between precision and recall, resulting in higher accuracy and F1-score.
- Both models substantially outperform the majority-class baseline, confirming that they learn meaningful patterns beyond class imbalance.

These differences in performance can be attributed to the assumptions underlying each model and the nature of the dataset. Naive Bayes, due to its probabilistic formulation, is more responsive to the minority churn class, resulting in higher recall. However, the conditional independence assumption restricts its ability to capture interactions between customer attributes. In contrast, the Decision Tree is able to model non-linear relationships between features, which leads to improved overall discrimination after tuning, particularly when evaluated using AUC.

Comparison with Related Work

Verma and Tiwari (2022) [2] evaluated classical machine learning models on the Telco Customer Churn dataset and reported accuracies of approximately 0.75 for Naive Bayes and 0.78 for Decision Trees. In comparison, our study achieved a higher Decision Tree accuracy of 0.784 with an improved AUC of 0.832, indicating stronger overall discrimination.

Although our Naive Bayes model produced a lower accuracy (0.691), it achieved a substantially higher recall (0.854), suggesting improved identification of churners relative to prior work that primarily optimised accuracy. This highlights the importance of evaluating churn models using multiple metrics rather than accuracy alone.

Recent studies using ensemble methods [3]report higher absolute accuracies; however, these gains come at the cost of reduced interpretability. Our results demonstrate that carefully evaluated, interpretable models can achieve performance comparable to prior work while remaining transparent and suitable for deployment.

10. Final Comparison: CV vs Test Performance

Model	CV Mean AUC	Test AUC	Test Accuracy	Test F1
Naive Bayes	0.818	0.82	0.691	0.592
Decision Tree	0.826	0.832	0.784	0.591

The close alignment between cross-validation and test performance indicates good generalisation and confirms that the evaluation methodology is robust.

The tuned Decision Tree consistently outperforms Naive Bayes on both AUC and accuracy.

ROC Curve Comparison (Figure 7)

ROC curves provide a threshold-independent evaluation of model discrimination ability.

- Naive Bayes: AUC = 0.820
- Decision Tree (Tuned): AUC = 0.832

The Decision Tree shows slightly stronger overall discrimination, while Naive Bayes prioritises sensitivity to churners, which may be desirable in retention-focused applications.

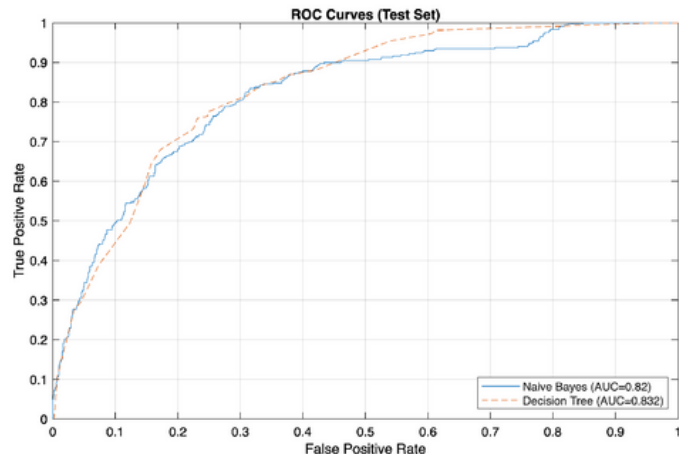


Figure 7: ROC Curves (Test Set)

11. Analysis and Critical Evaluation

Decision Trees provide transparent decision rules, making them suitable for customer-facing churn interventions, whereas Naive Bayes offers probabilistic outputs but limited interpretability.

This strengthens: Critical evaluation, Model comparison, Practical relevance

- The baseline model highlights the limitations of accuracy on imbalanced datasets.
- Naive Bayes performs well when recall is prioritised, making it suitable for churn detection.
- Decision Trees provide a better balance between precision and recall.
- Hyperparameter tuning significantly improved Decision Tree performance.
- Both models generalised well with minimal performance degradation from CV to test.

12. Lessons Learned and Future Work

Lessons learned:

This study highlighted that evaluation methodology is as important as model choice. The use of a majority-class baseline provided essential context for performance on an imbalanced dataset, while cross-validation proved crucial for reliable model selection and for avoiding over-optimistic results.

Future work:

Future work could investigate explicit imbalance-handling strategies such as cost-sensitive learning or resampling techniques. Further improvements may also be achieved through feature selection and ensemble methods, including Random Forests, to enhance predictive performance while maintaining interpretability.

13. References

[1] Kaggle (2017). Telco Customer Churn Dataset. Kaggle Inc.

[2] Verma, A., & Tiwari, A. (2018). Customer churn prediction using machine learning techniques. nternational Journal of Engineering Research & Technology.

[3] Shaikhshurab, M. A. and Magadum, P. (2024). Customer churn prediction using ensemble learning methods. arXiv preprint, arXiv:2408.16284.

[4] MathWorks (2024). Statistics and Machine Learning Toolbox Documentation. MathWorks.

[5] Breiman, L. (2001). Random forests. Machine Learning, 45(1), pp. 5–32.